USMAN KOUSAR

Lahore, Pakistan | +92-319-1688545 | usmankahloon076@gmail.com | linkedin.com/in/Usman-Kousar | github.com/Usman-Kousar

PROFESSIONAL SUMMARY

Final year computer science student specializing in AI, Machine Learning and Data Science, with hands-on experience building end-to-end machine learning pipelines — from dataset collection and feature engineering to model optimization and production-style reporting. Proficient in Python, Scikit-learn, TensorFlow, PyTorch, and Hugging Face Transformers, with practical experience across classification, regression, and NLP tasks.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++
- AI Frameworks & Libraries: NLTK, TF-IDF
- **ML Frameworks:** Scikit-learn, TensorFlow, PyTorch
- **ML Techniques:** Supervised & Unsupervised Learning, Classification, Regression, Feature Engineering, Hyperparameter Tuning, Cross-Validation
- **NLP & Text Processing:** TF-IDF Vectorization, Tokenization, NLTK, Sentiment Analysis, Text Classification
- **Model Evaluation:** AUC-ROC, F1-Score, Precision-Recall, Confusion Matrix, Performance Metrics
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Statistical Analysis
- **Tools & Platforms:** Jupyter Notebook, GitHub, VS Code
- **Collaboration:** Cross-functional Collaboration, Technical Documentation, Stakeholder Communication

PROFESSIONAL EXPERIENCE

Machine Learning Intern | *Developers Hub Corporation* March 2026 – May 2026

- Built and evaluated machine learning models end-to-end using Python, Scikit-learn, and Pandas, applying supervised learning algorithms to extract actionable insights from complex, high-dimensional datasets.
- Performed exploratory data analysis (EDA) to surface trends, patterns, and anomalies, directly informing feature engineering decisions and improving model signal quality for downstream classification and regression tasks.
- Collaborated with cross-functional engineering teams to design and ship automated data cleaning and preprocessing pipelines, standardizing input formats across 3+ core data sources and improving model reliability in production.
- Produced statistical performance reports and visual dashboards using Matplotlib and Seaborn, translating model evaluation metrics — including AUC-ROC, F1-Score, and Precision-Recall — into clear insights for non-technical stakeholders.
- Researched and documented emerging machine learning techniques, contributing to team knowledge-sharing and ongoing model optimization workflows.

MACHINE LEARNING PROJECTS

DDoS Attack Detection using Machine Learning | *Python , Scikit-learn , Pandas , NumPy , Matplotlib*

- Developed a machine learning-based DDoS attack detection system by preprocessing network traffic data and engineering features for binary attack classification.
- Trained and compared Logistic Regression, Decision Tree, Random Forest, Naive Bayes, KNN, and SVM models using Scikit-learn.
- Evaluated model performance using Accuracy, Precision, Recall, F1-Score, ROC Curve, and Confusion Matrix to identify the best-performing classifier.

Fake News Detection System | *Python, TF-IDF, NLP, NLTK, Scikit-learn*

- Built a production-style NLP classification pipeline — tokenization, TF-IDF vectorization, and multi-model benchmarking (Naive Bayes, SVM, Logistic Regression) — to detect misinformation in news text.
- Tuned decision boundaries and evaluated models on precision-recall trade-offs, informed by EDA on linguistic patterns and class distributions, to achieve high detection accuracy.

Solar Power Plant Forecasting & Efficiency Optimization | *Python, Pandas, NumPy, Matplotlib, Seaborn*

- Built a predictive analytics model covering 22 multi-unit inverters across 68,700+ rows, applying regression and feature engineering to model AC/DC efficiency ratios and flag hardware anomalies via Z-score analysis.
- Engineered features mapping thermal stress and generation trends to isolate peak operational windows, reaching ~95.85% average active output efficiency; visualized findings for operational decision-making.

EDUCATION

University of Management & Technology Expected July 2027

Bachelor of Science in Computer Science | Specialization: Artificial Intelligence & Data Science — CGPA: 3.50/4.00

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Science, Probability & Statistics, Database Management Systems

LEADERSHIP & ACTIVITIES

- **ICFCS – International Conference on Future Computing Systems, Organizing Committee Member (2024–2025):** Coordinated logistics and technical presentation tracks for an international AI and computing conference.